

D SANTHOSH KUMAR

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AI/ML Engineering student with expertise in Full-Stack Development and Artificial Intelligence, focused on designing and developing scalable, production-grade systems using Python, MERN stack, and deep learning frameworks, with experience in Java.

EDUCATION

Karunya University, Coimbatore, Tamilnadu - CGPA 8.13
Bachelor of Technology in Artificial Intelligence and Machine Learning

Sri Chaitanya Institute, Vijayawada, Andhra Pradesh Percentage: 95.7%
Higher Secondary Certificate Class XII — State Board (2023)

Sri Chaitanya School, Kurnool, Andhra Pradesh Percentage: 88%
Secondary School Certificate Class X — CBSE (2021)

Relevant Courses: Computer Architecture, Machine Learning, Programming Languages (Java, Python), Data Science, Data Structures, Information Retrieval, Computer Networks, Natural Language Processing.

PROJECTS

Visual Search Engine using VLM, Intel Unnati Internship Program, Karunya University

- Built a multimodal visual search system using OpenAI CLIP (ViT-B/32), scaling retrieval from 5,000 → 20,000 images with sub-second query latency via ChromaDB (HNSW indexing)
- Containerized the full pipeline using Docker, reducing deployment to a single build command with automated dataset extraction and indexing.

Automated Parking Management System (APMS), Deep Learning, Karunya University

- Built a real-time parking management system using deep learning for automated license plate and vehicle color recognition, achieving 89% detection accuracy
- Automated entry/exit monitoring, occupancy tracking, and billing via a Streamlit dashboard connected to a live database, reducing manual processing time by 12%
- Integrated multi-zone occupancy tracking, enabling real-time slot availability updates across the system.

Smart Oven Customer Support AI, LLM Fine-Tuning, Karunya University

- Fine-tuned a large language model using Unsloth for domain-specific customer support of a smart oven product, improving response accuracy and reducing hallucinations.
- Designed the instruction dataset, managed training with Transformers/Hugging Face, and evaluated model performance against baseline LLM outputs.

EXPERIENCE

Intel Unnati Industrial Training (2025)

- Engineered an end-to-end multimodal visual search system querying a 5,000-image COCO dataset via natural language and image inputs using OpenAI CLIP (ViT-B/32) for 512-dimensional semantic embeddings.
- Implemented vector similarity search with ChromaDB (cosine distance, HNSW indexing) and built a real-time Streamlit UI displaying results with latency reporting.
- Scaled the retrieval pipeline from 5,000 to 20,000 images through optimised batch indexing (128-image batches), maintaining sub-second query latency.
- Containerised the full application using Docker with automated dataset download, extraction, and ChromaDB indexing baked into the build pipeline for reproducible deployment.
- Curated and structured COCO train2017 image-caption metadata to produce clean, high-quality vector representations for robust VLM-based retrieval.

TECHNICAL SKILLS

Languages: Python · Java · JavaScript · HTML · CSS

Frameworks: React.js · Express.js · Node.js · Django · Flask · Streamlit

AI / ML: PyTorch · TensorFlow · TorchVision · OpenAI CLIP · Transformers · Unsloth · ChromaDB

Databases: MongoDB · SQL · ChromaDB · SQLite

Tools & Platforms: Docker · GitHub · Snowflake · Microsoft Office

Concepts: Vision-Language Models · LLM Fine-Tuning · Vector Search · Deep Learning · NLP

You can explore all above projects on my [GitHub](#).

CERTIFICATIONS

AI Foundation Associate Certificate – Oracle (2025)

- Completed foundational training in artificial intelligence.

SnowPro Associate: Platform Certification – Snowflake (2025)

- Successfully passed the SnowPro Associate exam, demonstrating proficiency in Snowflake platform fundamentals, cloud data warehousing, and platform operations.

Quantum Fundamentals – Qubitech (2026)

- Gained strong foundational exposure to quantum computing concepts including qubits, superposition, entanglement, quantum circuits, and real-world applications of quantum technologies.